Program No: Page No:

Date:

Source Code:

```
#Program 1 - INVOICE GENERATION
print("Theophilus Emmanuel.P")
print("12234")
print("12-B")
prods = int(input("Enter the number of products purchased: "))
pids, names, qtys, prices, discounts = [], [], [], [], []
bill=0

for i in range(prods):
    print(f"\n--- Product {i + 1} Details ---")
    name = input("Enter product name: ")
    pid = "PDT-" + name[0].upper() + str(i + 1)
    qty = int(input("Enter quantity: "))
    price = float(input("Enter price per item: "))

    if price>=200 and price<500:
        print("This product has a discount of 2 %")
        disc=price*2/100
    elif price>=500 and price<1000:
        print("This product has a discount of 5 %")
        disc=price*5/100
    elif price>=1000 and price<5000:
        print("This product has a discount of 8 %")
        disc=price*8/100
    elif price>=5000:
        print("This product has a discount of 10 %")
        disc=price*10/100
    else:
        disc=0
        print("No discount")

    pids.append(pid)
    names.append(name)
    qtys.append(qty)
    prices.append(price)
    discounts.append(disc)

print(f"{'FINAL INVOICE':^65}")
print("=" * 65)

print(f"{'Prod_ID':<10}{'Prod_Name':<15}{'Qty':<8}{'Price':<10}{'Discount':<10}
{'Total':<10}")
print("-" * 65)

for i in range(prods):
    total = prices[i] * qtys[i] - discounts[i]
    name_disp = names[i][:15]
```

```
print(f"{pids[i]:<10}{name_disp:<15}{qtys[i]:<8}{prices[i]:<10.2f}{discounts[i]
:<10.2f}{total:<10.2f}")
    bill += total

print("-" * 65)
print(f"{'TOTAL BILL AMOUNT':>50}: {bill:.2f}")
```

Output:

Getting input:

```
Theophilus Emmanuel.P
12234
12-B
Enter the number of products purchased: 5

--- Product 1 Details ---
Enter product name: Soap
Enter quantity: 3
Enter price per item: 20
No discount

--- Product 2 Details ---
Enter product name: Pen
Enter quantity: 10
Enter price per item: 10
No discount

--- Product 3 Details ---
Enter product name: Laptop
Enter quantity: 1
Enter price per item: 15000
This product has a discount of 10 %

--- Product 4 Details ---
Enter product name: Charger
Enter quantity: 2
Enter price per item: 600
This product has a discount of 5 %

--- Product 5 Details ---
Enter product name: Phone
Enter quantity: 3
Enter price per item: 150000
This product has a discount of 10 %
```

Program No: Page No:

Date:

Final output:

```
                          FINAL INVOICE
================================================================
Prod_ID   Prod_Name      Qty     Price     Discount  Total
----------------------------------------------------------------
PDT-S1    Soap           3       20.00     0.00      60.00
PDT-P2    Pen            10      10.00     0.00      100.00
PDT-L3    Laptop         1       15000.00  1500.00   13500.00
PDT-C4    Charger        2       600.00    30.00     1170.00
PDT-P5    Phone          3       150000.00 15000.00  435000.00
----------------------------------------------------------------
                                  TOTAL BILL AMOUNT: 449830.00
```

Result:

The program has been successfully executed.

Program No: Page No:

Date:

Source code:

```
#Program 2 - ARMSTRONG NUMBER
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
print("\tPrinting 3 digit, 4 digit & 5 digit Armstrong Numbers")
print("\t**************************************************\n")
for num in range(100,100001):
    armst=0
    s=str(num)
    for loop in s:
        digit=int(loop)
        power=len(s)
        armst=armst+digit**power

    if num==100:
        print("3 digit Armstrong Numbers:\n")
    elif num==1000:
        print("\n4 digit Armstrong Numbers:\n")
    elif num==10000:
        print("\n5 digit Armstrong Numbers:\n")

    if armst==num:
        print(num)
```

Output:

```
Theophilus Emmanuel P
12234
12-B
        Printing 3 digit, 4 digit & 5 digit Armstrong Numbers
        **************************************************

3 digit Armstrong Numbers:

153
370
371
407

4 digit Armstrong Numbers:

1634
8208
9474

5 digit Armstrong Numbers:

54748
92727
93084
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 3 - FIBONACCI SERIES
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
def fibonacci(n):
    a=0
    b=1
    if n<=0:
        print("Invalid Input !!!")
    elif n==1:
        print("Fibonacci Series:\n",a)
    else:
        print("Fibonacci Series:\n",a,b,end=" ")
        for loop in range(n-2):
            c=a+b
            print(c,end=" ")
            a=b
            b=c
num=int(input("Enter the number of Fibonacci Series to be generated: "))
fibonacci(num)
```

Output:

```
Theophilus Emmanuel P
12234
12-B
Enter the number of Fibonacci Series to be generated: 10
Fibonacci Series:
 0 1 1 2 3 5 8 13 21 34
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 4 - ATM OPERATIONS
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
Balance=120000
ch='Y'
AcNo=int(input("Enter your Account Number: "))
Name=input("Enter your Name: ")

def Dep():
    global Balance
    print("\t\t\tDEPOSIT")
    print("\t\t\t-------")
    print("Your Existing Balance is: ",Balance)
    depamt=int(input("Enter the amount to be deposited: "))
    Balance+=depamt
    print(f"\n\t\t\tAmount of Rs.{depamt} is Credited Successfully")
    return Balance

def With():
    global Balance
    print("\t\t\tWITHDRAWAL")
    print("\t\t\t----------")
    print("Your Existing Balance is: ",Balance)
    withamt=int(input("Enter the amount to be withdrawn: "))
    Balance-=withamt
    print(f"\n\t\t\tAmount of Rs.{withamt} is Debited Successfully")
    return Balance

def BalEnq():
    print("\n\t\tYour Account Number: ",AcNo)
    print("\t\tYour Name: ",Name)
    print("\n\t\t\tYour Account Balance is: ",Balance)

while ch=='Y' or ch=='y':
    print("\n\t\t\tWELCOME",Name)
    print("\n\t1.Deposit")
    print("\t2.Withdrawal")
    print("\t3.Balance Enquiry")
    print("\t4.Exit\n")
    choice=int(input("Choose your transaction: "))
    if choice<1 or choice>4:
        print("\n!!! CHOOSE A VALID TRANSACTION !!!\n")
    elif choice==1:
        print("\nAccount Number: ", AcNo)
        print("Name: ", Name)
        print("\n\tYour Account Balance is: ",Dep())
```

```
    elif choice==2:
        print("\nAccount Number: ", AcNo)
        print("Name: ", Name)
        print("\n\tYour Account Balance is: ",With())
    elif choice==3:
        BalEnq()
    elif choice==4:
        break
    ch=input("\nDo you want to continue transaction(Y/N)? ")
```

Output:

Getting values from user:

```
Theophilus Emmanuel P
12234
12-B
Enter your Account Number: 00100586361
Enter your Name: Isaac
```

Deposit:

```
                        WELCOME Isaac

          1.Deposit
          2.Withdrawal
          3.Balance Enquiry
          4.Exit

   Choose your transaction: 1

   Account Number:  100586361
   Name:  Isaac
                        DEPOSIT
                        -------
   Your Existing Balance is:  120000
   Enter the amount to be deposited: 50000

                        Amount of Rs.50000 is Credited Successfully

          Your Account Balance is:  170000

   Do you want to continue transaction(Y/N)? y
```

Program No: Page No:
Date:

Withdrawal:

```
                          WELCOME Isaac

          1.Deposit
          2.Withdrawal
          3.Balance Enquiry
          4.Exit

    Choose your transaction:  2

    Account Number:  100586361
    Name:  Isaac
                          WITHDRAWAL
                          ----------
    Your Existing Balance is:  170000
    Enter the amount to be withdrawn:  20000

                          Amount of Rs.20000 is Debited Successfully

          Your Account Balance is:  150000
```

Balance enquiry:

```
                        WELCOME Isaac

         1.Deposit
         2.Withdrawal
         3.Balance Enquiry
         4.Exit

  Choose your transaction:  3

                Your Account Number:  100586361
                Your Name:  Isaac

                       Your Account Balance is:  150000
```

Exit:

```
                          WELCOME Isaac

          1.Deposit
          2.Withdrawal
          3.Balance Enquiry
          4.Exit

    Choose your transaction:  4
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 5 - AREA OF SHAPES
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
print("\t\t\tAREA OF SHAPES")
print("\t\t\t--------------")
print("1.Area of Square")
print("2.Area of Rectangle")
print("3.Area of Triangle")
print("4.Area of Circle")
print("5.Exit")
choice=int(input("Enter your choice: "))
import math
def square(side):
    sqr=side*side
    print("The Area of Square: ",sqr)
def rectangle(length,breadth):
    rect=length*breadth
    print("The Area of Rectangle: ",rect)
def triangle(base,height):
    tri=0.5*base*height
    print("The Area of Triangle: ",tri)
def circle(radius):
    circ=math.pi*radius**2
    print("The Area of Circle: ",circ)

if choice==1:
    s=eval(input("Enter the value of Side: "))
    square(s)
elif choice==2:
    l=eval(input("Enter the value of Length: "))
    b=eval(input("Enter the value of Breadth: "))
    rectangle(l,b)
elif choice==3:
    bse=eval(input("Enter the value of Base: "))
    h=eval(input("Enter the value of Height: "))
    triangle(bse,h)
elif choice==4:
    r=eval(input("Enter the value of Radius: "))
    circle(r)
elif choice==5:
    pass
else:
    print("INVALID SELECTION!!!")
```

Program No: Page No:

Date:

Output:

Area of square:

```
Theophilus Emmanuel P
12234
12-B
                        AREA OF SHAPES
                        --------------
1.Area of Square
2.Area of Rectangle
3.Area of Triangle
4.Area of Circle
5.Exit
Enter your choice:  1
Enter the value of Side:  2
The Area of Square:  4
```

Area of rectangle:

```
Theophilus Emmanuel P
12234
12-B
                        AREA OF SHAPES
                        --------------
1.Area of Square
2.Area of Rectangle
3.Area of Triangle
4.Area of Circle
5.Exit
Enter your choice:  2
Enter the value of Length:  12
Enter the value of Breadth:  8
The Area of Rectangle:  96
```

Area of Triangle:

```
Theophilus Emmanuel P
12234
12-B
                        AREA OF SHAPES
                        --------------
1.Area of Square
2.Area of Rectangle
3.Area of Triangle
4.Area of Circle
5.Exit
Enter your choice:  3
Enter the value of Base:  12
Enter the value of Height:  5
The Area of Triangle:  30.0
```

Program No: Page No:

Date:

Area of circle:

```
Theophilus Emmanuel P
12234
12-B
                        AREA OF SHAPES
                        --------------
1.Area of Square
2.Area of Rectangle
3.Area of Triangle
4.Area of Circle
5.Exit
Enter your choice:  4
Enter the value of Radius:  5
The Area of Circle:  78.53981633974483
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 6 - RANDOM GENERATOR (DICE GAME)
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import random
print("\t\t\tDICE GAME")
print("\t\t\t*********")
moves=int(input("How many moves do you want to play? "))
for loop in range(moves):
    print("\nMove",loop+1)
    print("\tYour Turn")
    print("\t---------")
    key=input("Press any key to roll the dice (Press Enter to skip)... ")
    if key:
        print("You --> ",random.randint(1,6))
    else:
        print("Skipped...")
    print("\tComputer's Turn")
    print("\t---------------")
    print("Computer --> ",random.randint(1,6))
```

TO GOD BE THE GLORY
STANES SCHOOL
CBSE
SPIRIT
BODY MIND
COIMBATORE
2016
EXCELSA SEQUAR

Program No: Page No:

Date:

Output:

```
Theophilus Emmanuel P
12234
12-B
                        DICE GAME
                        *********
How many moves do you want to play?  6

Move 1
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  a
You -->  4
        Computer's Turn
        ---------------
Computer -->  5

Move 2
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  4
You -->  6
        Computer's Turn
        ---------------
Computer -->  5

Move 3
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  c
You -->  2
        Computer's Turn
        ---------------
Computer -->  4

Move 4
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  8
You -->  1
        Computer's Turn
        ---------------
Computer -->  5

Move 5
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  1
You -->  6
        Computer's Turn
        ---------------
Computer -->  4

Move 6
        Your Turn
        ---------
Press any key to roll the dice (Press Enter to skip)...  3
You -->  6
        Computer's Turn
        ---------------
Computer -->  4
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 7 - COUNTING VOWELS, CONSONANTS, UPPER CASE & LOWER CASE CHARACTERS
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
file=open("Quotes.txt","w+")
file.write('''Where there is righteousness in the heart, there is beauty in the
character.
When there is beauty in the character, there is harmony in the home.
When there is harmony in the home, there is an order in the nation.
When there is order in the nation, there is peace in the world.
- Dr. APJ ABDUL KALAM''')

file.seek(0)
data=file.read()
vowels=0
consonants=0
uppercs=0
lowercs=0
for letter in data:
    if letter.lower() in 'aeiou':
        vowels+=1
    if letter.lower() not in 'aeiou' and letter.isalpha():
        consonants+=1
    if letter.isupper():
        uppercs+=1
    if letter.islower():
        lowercs+=1
print(f"The no. of Vowels: {vowels}")
print(f"The no. of Consonants: {consonants}")
print(f"The no. of Upper Case letters: {uppercs}")
print(f"The no. of Lower Case letters: {lowercs}")
print(f"Total no. of Characters: {len(data)}")
file.close()
```

TO GOD BE THE GLORY
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Program No: Page No:

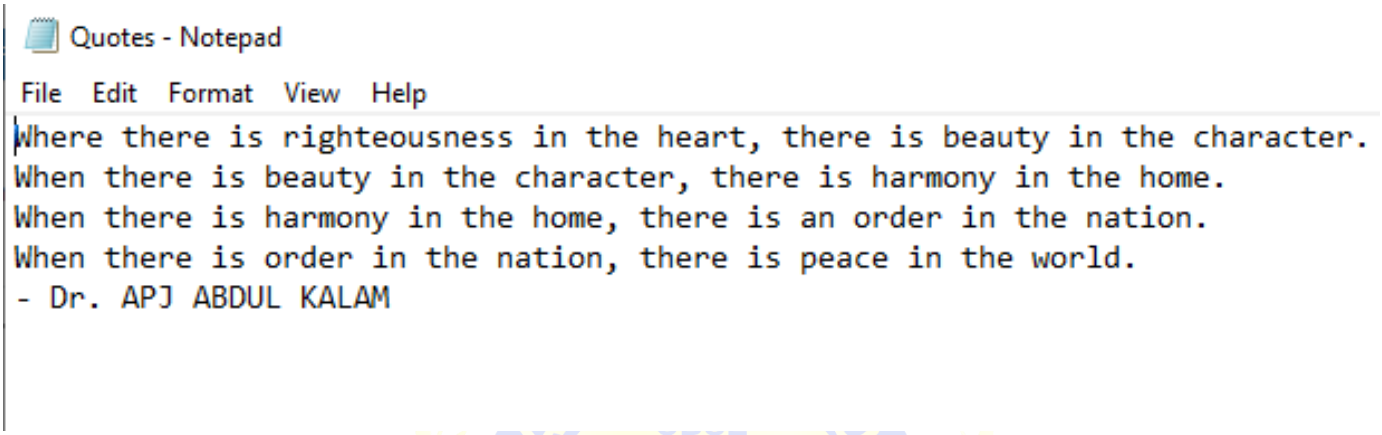
Date:

Output:

Python output:

```
Theophilus Emmanuel P
12234
12-B
The no. of Vowels: 92
The no. of Consonants: 140
The no. of Upper Case letters: 18
The no. of Lower Case letters: 214
Total no. of Characters: 299
```

Notepad output:



Result:

The program has been successfully executed

Program No: Page No:

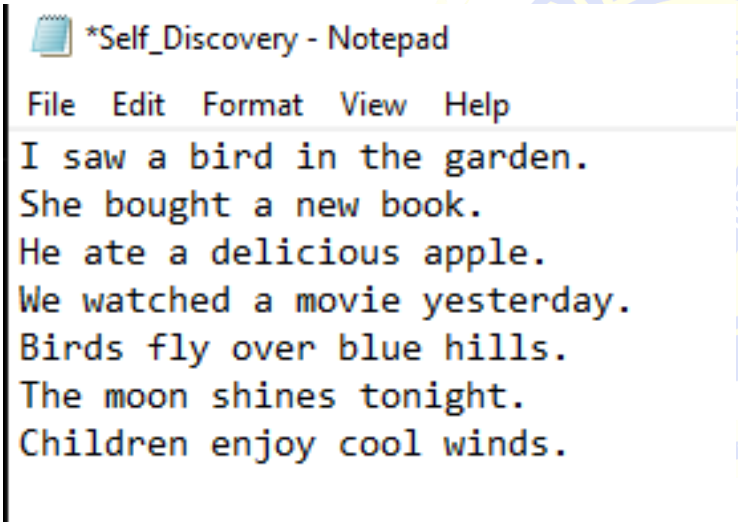
Date:

Source code:

```
#Program 8 - REMOVING LINES CONTAINING 'a'
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
rfile=open("Self_Discovery.txt","r+")
data=rfile.read()
print("The Original File Content:\n\n",data)
print()
with open("Self_Discovery.txt","w+") as wfile:
    for line in data.splitlines():
        if 'a' not in line:
            wfile.write(line)
            wfile.write("\n")
    wfile.seek(0)
    content=wfile.read()
    print("The file content after removing lines containing 'a':\n\n", content)
```

Output:

Before executing the code:



Program No: Page No:

Date:

Executing the code:

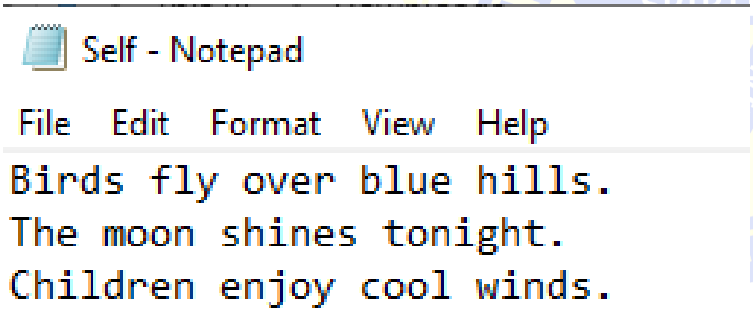
```
Theophilus Emmanuel P
12234
12-B
The Original File Content:

 I saw a bird in the garden.
She bought a new book.
He ate a delicious apple.
We watched a movie yesterday.
Birds fly over blue hills.
The moon shines tonight.
Children enjoy cool winds.

The file content after removing lines containing 'a':

 Birds fly over blue hills.
The moon shines tonight.
Children enjoy cool winds.
```

After executing the code:



Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

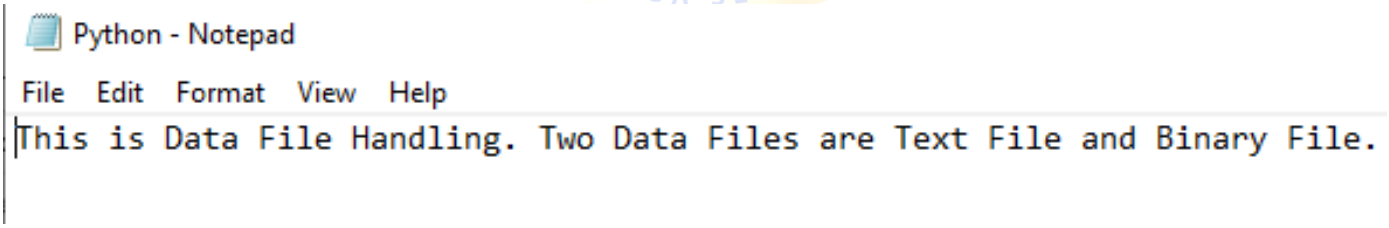
```
#Program 9 - WORDS SEPARATOR '#'
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
w_fle=open("Python.txt","w")
w_fle.write('''This is Data File Handling. Two Data Files are Text File and
Binary File.''')
w_fle.flush()
with open("Python.txt","r") as r_fle:
    sent=r_fle.read()
    print("Original File Content:")
    print(sent)
    print("Words separated by '#':")
    for loop in range(len(sent)):
        if sent[loop].isspace():
            print('#',end='')
        else:
            print(sent[loop],end='')
```

Output:

Python output:

```
Theophilus Emmanuel P
12234
12-B
Original File Content:
This is Data File Handling. Two Data Files are Text File and Binary File.
Words separated by '#':
This#is#Data#File#Handling.#Two#Data#Files#are#Text#File#and#Binary#File.
```

Notepad output:



Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 10 - SEARCHING IN A BINARY FILE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import pickle
students=[]
num=int(input("How many records you want to enter: "))
for loop in range(num):
    rno=int(input(f"Enter the Roll Number of the Student {loop+1}: "))
    name=input(f"Enter the Name of the Student {loop+1}: ")
    cls=int(input(f"Enter the Class of the Student {loop+1}: "))
    rec=[rno,name,cls]
    students.append(rec)
with open("Students_Data.dat","wb+") as f:
    pickle.dump(students,f)
    f.seek(0)
    ls=pickle.load(f)
    search=int(input("\nEnter the Roll Number of the Student you want to
search: "))
    for row in ls:
        if row[0]==search:
            print("\nRecord found...", row,sep="\n")
            break
    else:
        print("\nRecord Not Found")
```

Output:

Record found:

```
Theophilus Emmanuel P
12234
12-B
How many records you want to enter:  6
Enter the Roll Number of the Student 1:  12218
Enter the Name of the Student 1:  Johan
Enter the Class of the Student 1:  12
Enter the Roll Number of the Student 2:  12219
Enter the Name of the Student 2:  Kanish
Enter the Class of the Student 2:  12
Enter the Roll Number of the Student 3:  12212
Enter the Name of the Student 3:  Daniel
Enter the Class of the Student 3:  12
Enter the Roll Number of the Student 4:  12216
Enter the Name of the Student 4:  Isaac
Enter the Class of the Student 4:  12
Enter the Roll Number of the Student 5:  12234
Enter the Name of the Student 5:  Theophilus
Enter the Class of the Student 5:  12
Enter the Roll Number of the Student 6:  12237
Enter the Name of the Student 6:  Yashik
Enter the Class of the Student 6:  12

Enter the Roll Number of the Student you want to search:  12237

Record found...
[12237, 'Yashik', 12]
```

Program No: Page No:

Date:

Record not found:

```
Theophilus Emmanuel P
12234
12-B
How many records you want to enter:  6
Enter the Roll Number of the Student 1:  12218
Enter the Name of the Student 1:  Johan
Enter the Class of the Student 1:  12
Enter the Roll Number of the Student 2:  12219
Enter the Name of the Student 2:  Kanish
Enter the Class of the Student 2:  12
Enter the Roll Number of the Student 3:  12215
Enter the Name of the Student 3:  Isaac
Enter the Class of the Student 3:  12
Enter the Roll Number of the Student 4:  12212
Enter the Name of the Student 4:  Daniel
Enter the Class of the Student 4:  12
Enter the Roll Number of the Student 5:  12234
Enter the Name of the Student 5:  Theophilus
Enter the Class of the Student 5:  12
Enter the Roll Number of the Student 6:  12237
Enter the Name of the Student 6:  Yashik
Enter the Class of the Student 6:  12

Enter the Roll Number of the Student you want to

Record Not Found
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 11 - UPDATING BINARY FILE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import pickle
marklist=[]
num=int(input("How many records you want to enter: "))
for loop in range(num):
    rno=int(input(f"Enter the Roll Number of the Student {loop+1}: "))
    name=input(f"Enter the Name of the Student {loop+1}: ")
    mark=float(input(f"Enter the Mark of the Student {loop+1}: "))
    det=[rno,name,mark]
    marklist.append(det)
with open("Mark_List.dat","wb+") as bfile:
    pickle.dump(marklist,bfile)
    bfile.seek(0)
    mlst=pickle.load(bfile)
    print("Old Mark List:",mlst,sep="\n")
    srh=int(input("\nEnter the Roll Number of the Student whose mark is to be
updated: "))
    for rec in mlst:
        if rec[0]==srh:
            new=float(input(f"Enter the New Mark of the Student: "))
            rec[2]=new
            bfile.seek(0)
            pickle.dump(mlst,bfile)
            bfile.seek(0)
            updlst=pickle.load(bfile)
            print("Mark List Updated...")
            print(updlst)
            break
    else:
        print("Record Not Found...")
```

Program No: Page No:

Date:

Output:

Getting input:

```
Theophilus Emmanuel P
12234
12-B
How many records you want to enter:  5
Enter the Roll Number of the Student 1:  12218
Enter the Name of the Student 1:  Johan
Enter the Mark of the Student 1:  55
Enter the Roll Number of the Student 2:  12219
Enter the Name of the Student 2:  Kanish
Enter the Mark of the Student 2:  68
Enter the Roll Number of the Student 3:  12212
Enter the Name of the Student 3:  Daniel
Enter the Mark of the Student 3:  99
Enter the Roll Number of the Student 4:  12215
Enter the Name of the Student 4:  Isaac
Enter the Mark of the Student 4:  100
Enter the Roll Number of the Student 5:  12234
Enter the Name of the Student 5:  Theophilus
Enter the Mark of the Student 5:  99
```

Record found:

```
Enter the Roll Number of the Student whose mark is to be updated:  12212
Enter the New Mark of the Student:  98
Mark List Updated...
[[12218, 'Johan', 55.0], [12219, 'Kanish', 68.0], [12212, 'Daniel', 98.0], [12215, 'Isaac', 100.0], [12234, 'Theophilus', 99.0]]
```

Record not found:

```
Enter the Roll Number of the Student whose mark is to be updated:  12234
Record Not Found...
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 12- CREATING INVENTORY - CSV FILE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import csv
invntry={}
stkno=[]
stkname=[]
qty=[]
choice='Y'
print("\t\t\tINVENTORY")
print("\t\t\t*********")
while choice=='Y' or choice=='y':
    sno=input("Enter the Stock Serial Number: ")
    name=input("Enter the Stock Name: ")
    q=input("Enter the Quantity: ")
    stkno.append(sno)
    stkname.append(name)
    qty.append(q)
    choice=input("Do you want to continue?(Y/N): ")
invntry.update(StockNo=stkno,StockName=stkname,Quantity=qty)

with open("Inventory.csv","w+", newline="\n") as cfile:
    w=csv.writer(cfile)
    w.writerow(invntry.keys())
    for loop in range(len(stkno)):

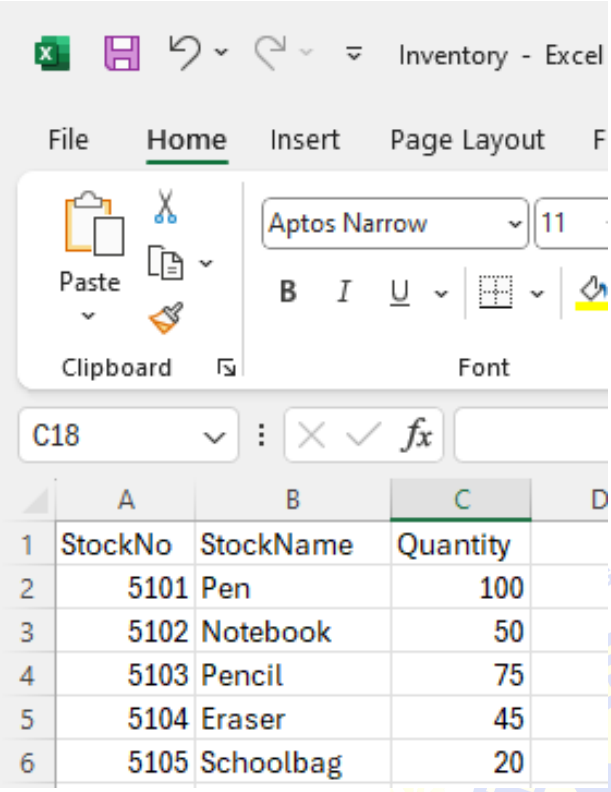
w.writerow([invntry['StockNo'][loop],invntry['StockName'][loop],invntry['Quanti
ty'][loop]])
    r=csv.reader(cfile)
    cfile.seek(0)
    for row in r:
        print(row)
```

Program No: Page No:

Date:

Output:

In Excel:



	A	B	C	D
1	StockNo	StockName	Quantity	
2	5101	Pen	100	
3	5102	Notebook	50	
4	5103	Pencil	75	
5	5104	Eraser	45	
6	5105	Schoolbag	20	

In Python:

```
Theophilus Emmanuel P
12234
12-B
                        INVENTORY
                        *********
Enter the Stock Serial Number:  5101
Enter the Stock Name:  Pen
Enter the Quantity:  100
Do you want to continue?(Y/N):  y
Enter the Stock Serial Number:  5102
Enter the Stock Name:  Notebook
Enter the Quantity:  50
Do you want to continue?(Y/N):  y
Enter the Stock Serial Number:  5103
Enter the Stock Name:  Pencil
Enter the Quantity:  75
Do you want to continue?(Y/N):  y
Enter the Stock Serial Number:  5104
Enter the Stock Name:  Eraser
Enter the Quantity:  45
Do you want to continue?(Y/N):  y
Enter the Stock Serial Number:  5105
Enter the Stock Name:  Schoolbag
Enter the Quantity:  20
Do you want to continue?(Y/N):  n
['StockNo', 'StockName', 'Quantity']
['5101', 'Pen', '100']
['5102', 'Notebook', '50']
['5103', 'Pencil', '75']
['5104', 'Eraser', '45']
['5105', 'Schoolbag', '20']
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 13 - CREATING USER-ID & PASSWORD USING CSV FILE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import csv
CSV_FILE = 'users.csv'
def create_csv():
    n = int(input("Enter number of users to add: "))
    with open(CSV_FILE, 'w', newline='') as f:
        writer = csv.writer(f)
        writer.writerow(['User_ID', 'Password'])
        for i in range(n):
            user_id = input(f"Enter User-ID for user {i+1}: ")
            password = input(f"Enter Password for {user_id}: ")
            writer.writerow([user_id, password])
    print("\nCSV file created successfully!\n")

def display_csv():
    print("\nUser-ID and Passwords stored in CSV file:")
    with open(CSV_FILE, 'r', newline='') as f:
        reader = csv.reader(f)
        for row in reader:
            print(row)

def search_password():
    uid = input("\nEnter User-ID to search: ")
    found = False
    with open(CSV_FILE, 'r', newline='') as f:
        reader = csv.reader(f)
        for row in reader:
            if row[0] == uid:
                print(f"Password for '{uid}' is: {row[1]}")
                found = True
                break
    if not found:
        print(f"User-ID '{uid}' not found in file.")

while True:
    print("1. Create CSV File")
    print("2. Display All Records")
    print("3. Search Password by User-ID")
    print("4. Exit")
    choice = input("Enter your choice (1-4): ")

    if choice == '1':
        create_csv()
    elif choice == '2':
        display_csv()
    elif choice == '3':
        search_password()
```

```
    elif choice == '4':
        print("Program terminated.")
        break
    else:
        print("Invalid choice! Please try again.")
```

Output:

Creating csv file:

```
Theophilus Emmanuel P
12234
12-B
1. Create CSV File
2. Display All Records
3. Search Password by User-ID
4. Exit
Enter your choice (1-4):  1
Enter number of users to add:  5
Enter User-ID for user 1:  user101
Enter Password for user101:  Theo@123
Enter User-ID for user 2:  user102
Enter Password for user102:  Music#456
Enter User-ID for user 3:  user103
Enter Password for user103:  Study@789
Enter User-ID for user 4:  user104
Enter Password for user104:  Piano#234
Enter User-ID for user 5:  user105
Enter Password for user105:  Class12@567

CSV file created successfully!
```

Displaying all records:

```
    1. Create CSV File
    2. Display All Records
    3. Search Password by User-ID
    4. Exit
    Enter your choice (1-4):  2

    User-ID and Passwords stored in CSV file:
    ['User_ID', 'Password']
    ['user101', 'Theo@123']
    ['user102', 'Music#456']
    ['user103', 'Study@789']
    ['user104', 'Piano#234']
    ['user105', 'Class12@567']
```

Program No: Page No:

Date:

Searching for password:

```
1. Create CSV File
2. Display All Records
3. Search Password by User-ID
4. Exit
Enter your choice (1-4):  3

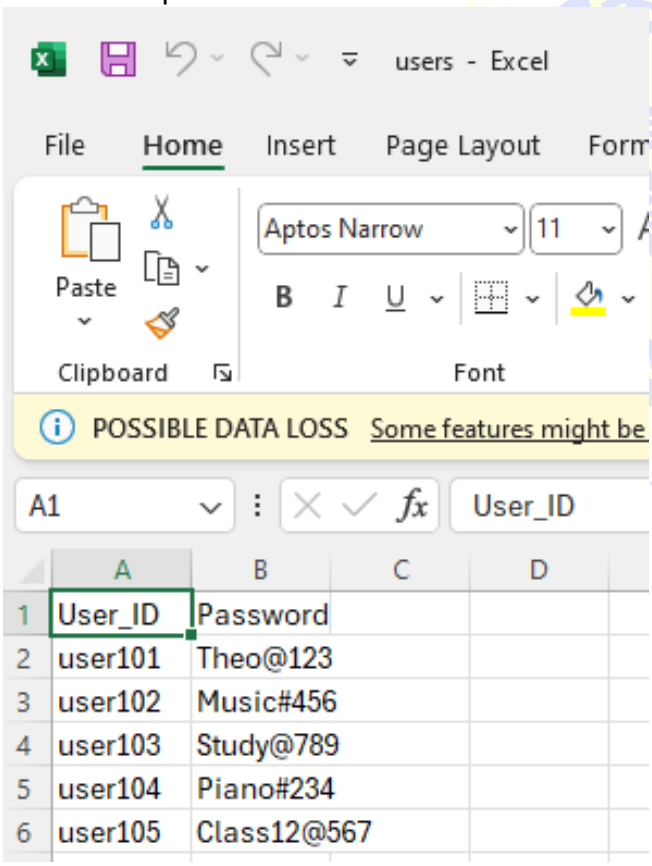
Enter User-ID to search:  user103
Password for 'user103' is: Study@789
```

If the password is not found:

```
1. Create CSV File
2. Display All Records
3. Search Password by User-ID
4. Exit
Enter your choice (1-4):  3

Enter User-ID to search:  user106
User-ID 'user106' not found in file.
```

Excel output:



	A	B	C	D
1	User_ID	Password		
2	user101	Theo@123		
3	user102	Music#456		
4	user103	Study@789		
5	user104	Piano#234		
6	user105	Class12@567		

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 14 - STACK OPERATIONS
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
def Push(stk, elem):
    stk.append(elem)
    top=len(stk)-1

def Pop(stk):
    if stk==[]:
        return 'Underflow'
    else:
        del_elem=stk.pop()
        if len(stk)==0:
            top=None
        else:
            top=len(stk)-1
        return del_elem

def Peek(stk):
    if stk==[]:
        return 'Underflow'
    else:
        top=len(stk)-1
        return stk[top]

def Display(stk):
    if stk==[]:
        print("\n\tStack is Empty !")
    else:
        top=len(stk)-1
        print("\nStack contains:")
        print("\t",stk[top],"<- Top element")
        for loop in range(top-1, -1, -1):
            print("\t",stk[loop])

Stack=[]
top=None
while True:
    print("\n\t\t\tSTACK OPERATIONS")
    print("\t\t\t***************")
    print("1.Push")
    print("2.Pop")
    print("3.Peek")
    print("4.Display")
    print("5.Exit")
    ch=int(input("Enter your choice: "))
    if ch<1 or ch>5:
        print("\n!!! INVALID SELECTION !!! Choose from 1 to 5")
```

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BODY MIND
COIMBATORE
2016
EXCELSA SEQUAR

```
    elif ch==1:
        Element=int(input("Enter the element: "))
        Push(Stack, Element)

    elif ch==2:
        d_elem=Pop(Stack)
        if d_elem=='Underflow':
            print("\n\tUnderflow ! The Stack is Empty")
        else:
            print("\n\tThe Popped element is: ", d_elem)
    elif ch==3:
        top_elem=Peek(Stack)
        if top_elem=='Underflow':
            print("\n\tUnderflow ! The Stack is Empty")
        else:
            print("\n\tThe Topmost element is: ", top_elem)
    elif ch==4:
        Display(Stack)
    elif ch==5:
        break
```

Output:

Push operation:

```
Theophilus Emmanuel P
12234
12-B

                        STACK OPERATIONS
                        ****************

1.Push
2.Pop
3.Peek
4.Display
5.Exit
Enter your choice:  1
Enter the element:  10

                        STACK OPERATIONS
                        ****************

1.Push
2.Pop
3.Peek
4.Display
5.Exit
Enter your choice:  1
Enter the element:  20

                        STACK OPERATIONS
                        ****************

1.Push
2.Pop
3.Peek
4.Display
5.Exit
Enter your choice:  1
Enter the element:  30
```

Program No: Page No:

Date:

Pop operation:

```
                        STACK OPERATIONS
                        ****************
  1.Push
  2.Pop
  3.Peek
  4.Display
  5.Exit
  Enter your choice:  2

          The Popped element is:  30
```

Peek operation:

```
                         STACK OPERATIONS
                         ****************
    1.Push
    2.Pop
    3.Peek
    4.Display
    5.Exit
    Enter your choice:  3

           The Topmost element is:  20
```

Display operation:

```
                        STACK OPERATIONS
                        ****************
  1.Push
  2.Pop
  3.Peek
  4.Display
  5.Exit
  Enter your choice:  4

  Stack contains:
          20 <- Top element
          10
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 15 - LINEAR SEARCH
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
def LinearSearch(lst,key):
    found=None
    for loop in range(len(lst)):
        if lst[loop]==key:
            found=lst.index(lst[loop])
    if found==None:
        return -1
    else:
        return found

ls=eval(input("Enter the elements for the list: "))
opt='Y'
if type(ls)==tuple:
    print("\t!!! ENTER THE ELEMENTS IN SQUARE BRACKETS[] !!!")
else:
    while opt=='Y' or opt=='y':
        search=int(input("\nEnter the element to be searched: "))
        ind=LinearSearch(ls,search)
        if ind==-1:
            print("\n\tSorry! Element is not in the List")
        else:
            print(f"\n\tThe Element {search} is found at Index Position {ind}")
        opt=input("\nDo you want to search any other elements? (Y/N): ")
```

Output:

Element found:

```
Theophilus Emmanuel P
12234
12-B
Enter the elements for the list:  [10,20,30,40,50,60,70]

Enter the element to be searched:  40

        The Element 40 is found at Index Position 3
```

Element not found:

```
Enter the element to be searched:  80

        Sorry! Element is not in the List

Do you want to search any other elements? (Y/N):  n
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 16 - BINARY SEARCH
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
def BinarySearch(lst,key):
    beg=0
    last=len(ls)-1
    while beg<=last:
        mid=(beg+last)//2
        if search==ls[mid]:
            return mid
        elif search>ls[mid]:
            beg=mid+1
        elif search<ls[mid]:
            last=mid-1
    else:
        return -1

ls=eval(input("Enter the elements for the list: "))
opt='Y'
if type(ls)==tuple:
    print("\t!!! ENTER THE ELEMENTS IN SQUARE BRACKETS[] !!!")
else:
    while opt=='Y' or opt=='y':
        ls.sort()
        print("The Sorted List: ", ls)
        search=int(input("\nEnter the element to be searched: "))
        ind=BinarySearch(ls,search)
        if ind==-1:
            print("\n\tSorry! Element is not in the List")
        else:
            print(f"\n\tThe Element {search} is found at Index Position {ind}")
        opt=input("\nDo you want to search any other elements? (Y/N): ")
```

Program No: Page No:

Date:

Output:

Element is found:

```
Theophilus Emmanuel P
12234
12-B
Enter the elements for the list:  [10,30,50,20,60,70]
The Sorted List:  [10, 20, 30, 50, 60, 70]

Enter the element to be searched:  30

        The Element 30 is found at Index Position 2
```

Element not found:

```
Do you want to search any other elements? (Y/N):  y
The Sorted List:  [10, 20, 30, 50, 60, 70]

Enter the element to be searched:  40

        Sorry! Element is not in the List

Do you want to search any other elements? (Y/N):  n
```

Result:

The program has been successfully executed

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Program No: Page No:

Date:

Source code:

```
#Program 17 - GENERATING CREATE, INSERT QUERIES - PYTHON MYSQL INTERFACE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import mysql.connector as sql
conn=sql.connect(host="localhost", user="root", password="mysql123",
database="Class12")
if conn.is_connected():
    print("Connection Established Successfully... Process Started...")
crsr=conn.cursor()
create='''CREATE TABLE Student
(RollNo VARCHAR(20), Name CHAR(20), DOB DATE, PhoneNo BIGINT, Location
VARCHAR(30))'''
crsr.execute(create)
conn.commit()
print("Student Table is Created Successfully...")
print("\tEnter Data for the Table Student:")
rec=int(input("How many records do you want to insert in the Table Student? "))
print()
for loop in range(rec):
    rno=input(f"Enter the Roll Number of the Student {loop+1}: ")
    name=input(f"Enter the Name of the Student {loop+1}: ")
    dob=int(input(f"Enter the DOB of the Student {loop+1} in YYYYMMDD format:
"))
    phone=int(input(f"Enter the Phone Number of the Student {loop+1}: "))
    loc=input(f"Enter the Location of the Student {loop+1}: ")
    crsr.execute(f"INSERT INTO Student
VALUES('{rno}','{name}',{dob},{phone},'{loc}')")
    conn.commit()
    print(f"\t{loop+1} Records Inserted Successfully...\n")
conn.close()
```

Program No: Page No:

Date:

Output:

Python output:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...
Student Table is Created Successfully...
        Enter Data for the Table Student:
How many records do you want to insert in the Table Student?  5

Enter the Roll Number of the Student 1:  12218
Enter the Name of the Student 1:  Johan
Enter the DOB of the Student 1 in YYYYMMDD format:  20091004
Enter the Phone Number of the Student 1:  9003087272
Enter the Location of the Student 1:  Hopes college
        1 Records Inserted Successfully...

Enter the Roll Number of the Student 2:  12219
Enter the Name of the Student 2:  Kanish
Enter the DOB of the Student 2 in YYYYMMDD format:  20090905
Enter the Phone Number of the Student 2:  9952489509
Enter the Location of the Student 2:  Avinashi Road
        2 Records Inserted Successfully...

Enter the Roll Number of the Student 3:  12212
Enter the Name of the Student 3:  Daniel
Enter the DOB of the Student 3 in YYYYMMDD format:  20090709
Enter the Phone Number of the Student 3:  9958761029
Enter the Location of the Student 3:  Gandhipuram
        3 Records Inserted Successfully...


Enter the Roll Number of the Student 4:  12234
Enter the Name of the Student 4:  Theophilus
Enter the DOB of the Student 4 in YYYYMMDD format:  20090508
Enter the Phone Number of the Student 4:  9840541290
Enter the Location of the Student 4:  Ram nagar
        4 Records Inserted Successfully...

Enter the Roll Number of the Student 5:  12237
Enter the Name of the Student 5:  Yashik
Enter the DOB of the Student 5 in YYYYMMDD format:  20090512
Enter the Phone Number of the Student 5:  9443348599
Enter the Location of the Student 5:  Ramanathapuram
        5 Records Inserted Successfully...
```

Program No: Page No:

Date:

SQL output:

```
mysql> desc student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RollNo   | varchar(20) | YES  |     | NULL    |       |
| Name     | char(20)    | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| PhoneNo  | bigint      | YES  |     | NULL    |       |
| Location | varchar(30) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> Select * from student;
+--------+------------+------------+------------+----------------+
| RollNo | Name       | DOB        | PhoneNo    | Location       |
+--------+------------+------------+------------+----------------+
| 12218  | Johan      | 2009-10-04 | 9003087272 | Hopes college  |
| 12219  | Kanish     | 2009-09-05 | 9952489509 | Avinashi Road  |
| 12212  | Daniel     | 2009-07-09 | 9958761029 | Gandhipuram    |
| 12234  | Theophilus | 2009-05-08 | 9840541290 | Ram nagar      |
| 12237  | Yashik     | 2009-05-12 | 9443348599 | Ramanathapuram |
+--------+------------+------------+------------+----------------+
5 rows in set (0.00 sec)
```

Result:

The program has been successfully executed

Program No: Page No:

Date:

Source code:

```
#Program 18 - GENERATING UPDATE, DELETE QUERIES - PYTHON MYSQL INTERFACE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import mysql.connector as sql
conn=sql.connect(host="localhost", user="root", password="mysql123",
database="Class12")
if conn.is_connected():
    print("Connection Established Successfully... Process Started...\n")
crsr=conn.cursor()
print("\tUPDATE STUDENT PHONE NUMBER...")
ch=input("\nDo You Want To Update Any Record?(Y/N): ")
while ch=='Y' or ch=='y':
    name=input("\nEnter the Name of the Student: ")
    phone=int(input(f"Enter the New Phone Number of {name}: "))
    crsr.execute(f"UPDATE Student SET PhoneNo={phone} WHERE Name='{name}'")
    conn.commit()
    print(f"\n\t{name} Phone Number is Updated to {phone} Successfully...")
    ch=input("\nDo You Want To Update Any Other Record?(Y/N): ")

print("\n\tDELETE STUDENT RECORD...")
ch1=input("\nDo You Want To Delete Any Record?(Y/N): ")
while ch1=='Y' or ch1=='y':
    rno=input("\nEnter the Roll Number of the Student: ")
    crsr.execute(f"DELETE FROM Student WHERE RollNo='{rno}'")
    conn.commit()
    print(f"\n\t{rno} Record is Deleted Successfully...")
    ch1=input("\nDo You Want To Delete Any Other Record?(Y/N): ")
conn.close()
```

Output:

Before execution of code:

```
mysql> Select * from student;
+--------+------------+------------+------------+----------------+
| RollNo | Name       | DOB        | PhoneNo    | Location       |
+--------+------------+------------+------------+----------------+
| 12218  | Johan      | 2009-10-04 | 9003087272 | Hopes college  |
| 12219  | Kanish     | 2009-09-05 | 9952489509 | Avinashi Road  |
| 12212  | Daniel     | 2009-07-09 | 9958761029 | Gandhipuram    |
| 12234  | Theophilus | 2009-05-08 | 9840541290 | Ram nagar      |
| 12237  | Yashik     | 2009-05-12 | 9443348599 | Ramanathapuram |
+--------+------------+------------+------------+----------------+
```

Program No: Page No:

Date:

Python output:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...

        UPDATE STUDENT PHONE NUMBER...

Do You Want To Update Any Record?(Y/N):  y

Enter the Name of the Student:  Johan
Enter the New Phone Number of Johan:  9080425292

        Johan Phone Number is Updated to 9080425292 Successfully...

Do You Want To Update Any Other Record?(Y/N):  n

        DELETE STUDENT RECORD...

Do You Want To Delete Any Record?(Y/N):  y

Enter the Roll Number of the Student:  12234

        12234 Record is Deleted Successfully...

Do You Want To Delete Any Other Record?(Y/N):  n
```

After execution of code:

```
mysql> Select * from student;
+--------+--------+------------+------------+----------------+
| RollNo | Name   | DOB        | PhoneNo    | Location       |
+--------+--------+------------+------------+----------------+
| 12218  | Johan  | 2009-10-04 | 9080425292 | Hopes college  |
| 12219  | Kanish | 2009-09-05 | 9952489509 | Avinashi Road  |
| 12212  | Daniel | 2009-07-09 | 9958761029 | Gandhipuram    |
| 12237  | Yashik | 2009-05-12 | 9443348599 | Ramanathapuram |
+--------+--------+------------+------------+----------------+
4 rows in set (0.00 sec)

mysql>
```

Result:

The program has been executed successfully

Program No: Page No:

Date:

Source code:

```
#Program 19 - GENERATING SELECT QUERY - PYTHON MYSQL INTERFACE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import mysql.connector as sql
conn=sql.connect(host="localhost", user="root", password="mysql123",
database="Class12")
if conn.is_connected():
    print("Connection Established Successfully... Process Started...\n")
crsr=conn.cursor()
select="SELECT * FROM Student ORDER BY DOB"
crsr.execute(select)
table=crsr.fetchall()
count=crsr.rowcount
print(f"The Number of Records retrieved are: {count}")
print("The Records in the Table Student:")
for row in table:
    print(row)

ch=input("\nDo You Want To Retrieve Any Specific Number Of Records?(Y/N): ")
while ch=='Y' or ch=='y':
    rec=int(input("\nHow many Records do you need? "))
    crsr.execute(select)
    data=crsr.fetchmany(rec)
    count2=crsr.rowcount
    print(f"\nThe Number of Records now retrieved are: {count2}")
    print(f"{rec} Records from the Table Student:")
    for record in data:
        print(record)
    ch=False
conn.close()
```

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Program No: Page No:

Date:

Output:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...

The Number of Records retrieved are: 4
The Records in the Table Student:
('12237', 'Yashik', datetime.date(2009, 5, 12), 9443348599, 'Ramanathapuram')
('12212', 'Daniel', datetime.date(2009, 7, 9), 9958761029, 'Gandhipuram')
('12219', 'Kanish', datetime.date(2009, 9, 5), 9952489509, 'Avinashi Road')
('12218', 'Johan', datetime.date(2009, 10, 4), 9080425292, 'Hopes college')

Do You Want To Retrieve Any Specific Number Of Records?(Y/N):  y

How many Records do you need?  2

The Number of Records now retrieved are: 2
2 Records from the Table Student:
('12237', 'Yashik', datetime.date(2009, 5, 12), 9443348599, 'Ramanathapuram')
('12212', 'Daniel', datetime.date(2009, 7, 9), 9958761029, 'Gandhipuram')
```

Result:

The program has been executed successfully

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Program No: Page No:

Date:

Source code:

```
#Program 20 - GENERATING ALTER QUERY - PYTHON SQL INTERFACE
print("Theophilus Emmanuel P")
print("12234")
print("12-B")
import mysql.connector as sql
conn=sql.connect(host="localhost", user="root", password="mysql123",
database="Class12")
if conn.is_connected():
    print("Connection Established Successfully... Process Started...")
crsr=conn.cursor()
print("\n\tAlter Operations for the Table Student")
print("\n1.Add New Column")
print("2.Modify Existing Column")
print("3.Rename Column")
print("4.Exit")
ch=int(input("Select The Operation You Want To Perform: "))
if ch==1:
    newcol=input("\nEnter the New Column Name: ")
    newdat=input(f"Enter the DataType with Size(if any) for {newcol}: ")
    crsr.execute(f"ALTER TABLE Student ADD {newcol} {newdat}")
    conn.commit()
    print("\n\tAdded a New Column Successfully...")
elif ch==2:
    col=input("\nEnter the Column Name to be Modified: ")
    chdat=input(f"Enter the New DataType with New Size(if any) for {col}: ")
    crsr.execute(f"ALTER TABLE Student MODIFY {col} {chdat}")
    conn.commit()
    print(f"\n\tModified {col} Successfully...")
elif ch==3:
    oldname=input("\nEnter the Column Name to be Renamed: ")
    newname=input(f"Enter the New Column Name with New Size(if any) for
{oldname}: ")
    crsr.execute(f"ALTER TABLE Student CHANGE {oldname} {newname}")
    conn.commit()
    print(f"\n\tRenamed {oldname} Successfully...")
elif ch==4:
    pass
else:
    print("!!! CHOOSE A VALID OPERATION !!!")
conn.close()
```

Program No: Page No:

Date:

Output:

Adding a column:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...

        Alter Operations for the Table Student

1.Add New Column
2.Modify Existing Column
3.Rename Column
4.Exit
Select The Operation You Want To Perform:  1

Enter the New Column Name:  class
Enter the DataType with Size(if any) for class:  int

        Added a New Column Successfully...
```

Modify Existing column:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...

        Alter Operations for the Table Student

1.Add New Column
2.Modify Existing Column
3.Rename Column
4.Exit
Select The Operation You Want To Perform:  2

Enter the Column Name to be Modified:  NAme
Enter the New DataType with New Size(if any) for NAme:  Varchar(20)

        Modified NAme Successfully...
```

Rename column:

```
Theophilus Emmanuel P
12234
12-B
Connection Established Successfully... Process Started...

        Alter Operations for the Table Student

1.Add New Column
2.Modify Existing Column
3.Rename Column
4.Exit
Select The Operation You Want To Perform:  3

Enter the Column Name to be Renamed:  Rollno
Enter the New Column Name with New Size(if any) for Rollno:  Rno INT

        Renamed Rollno Successfully...
```

Program No: Page No:

Date:

SQL output before execution:

```
mysql> desc student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RollNo   | varchar(20) | YES  |     | NULL    |       |
| Name     | char(20)    | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| PhoneNo  | bigint      | YES  |     | NULL    |       |
| Location | varchar(30) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> Select * from student;
+--------+--------+------------+------------+----------------+
| RollNo | Name   | DOB        | PhoneNo    | Location       |
+--------+--------+------------+------------+----------------+
| 12218  | Johan  | 2009-10-04 | 9080425292 | Hopes college  |
| 12219  | Kanish | 2009-09-05 | 9952489509 | Avinashi Road  |
| 12212  | Daniel | 2009-07-09 | 9958761029 | Gandhipuram    |
| 12237  | Yashik | 2009-05-12 | 9443348599 | Ramanathapuram |
+--------+--------+------------+------------+----------------+
4 rows in set (0.00 sec)
```

SQL output after execution:

```
mysql> desc student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| Rno      | int         | YES  |     | NULL    |       |
| NAme     | varchar(20) | YES  |     | NULL    |       |
| DOB      | date        | YES  |     | NULL    |       |
| PhoneNo  | bigint      | YES  |     | NULL    |       |
| Location | varchar(30) | YES  |     | NULL    |       |
| class    | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.04 sec)

mysql> Select * from student;
+-------+--------+------------+------------+----------------+-------+
| Rno   | NAme   | DOB        | PhoneNo    | Location       | class |
+-------+--------+------------+------------+----------------+-------+
| 12218 | Johan  | 2009-10-04 | 9080425292 | Hopes college  |  NULL |
| 12219 | Kanish | 2009-09-05 | 9952489509 | Avinashi Road  |  NULL |
| 12212 | Daniel | 2009-07-09 | 9958761029 | Gandhipuram    |  NULL |
| 12237 | Yashik | 2009-05-12 | 9443348599 | Ramanathapuram |  NULL |
+-------+--------+------------+------------+----------------+-------+
4 rows in set (0.00 sec)
```

Result:

The program has been successfully executed